

**Climate Warming Trends and Implications
for Red River Ski Area Operations:
A 37-Year Time Series Analysis**

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Time Series Analysis

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1 Data and Context

This analysis examines daily temperature records from the Red River pass weather station in Red River, New Mexico spanning January 1989 through November 2025. The dataset consists of 13,455 daily observations aggregated into 443 monthly records, tracking average temperature (TAVG), maximum temperature (TMAX), minimum temperature (TMIN), precipitation (PRCP), and snow depth (SNWD). The data was acquired from NOAA's public climate database and directly measures conditions affecting Red River Ski Area operations. The central business question examines how climate warming affects ski resort operations, specifically whether rising temperatures have reduced optimal snowmaking days during the ski season. This question impacts investment in snowmaking equipment, staffing, marketing, and long-term planning. To support the analysis, a binary variable identifies snowmaking days (temperature below 28°F), and the dataset is split into Early Era (1989-2006) and Recent Era (2007-Present) for comparative measurement of climate change impacts.

2 Methods and Time-Series Approach

The analytical approach employed a five-phase methodology to extract meaningful patterns from the temperature record. First, daily observations were converted into monthly averages using PivotTables, with missing data from July 2002 through August 2003 addressed using seasonal average imputation to preserve seasonal structure. Second, moving averages were created using both 3-month (smoothing short-term fluctuations while preserving seasonal transitions) and 12-month (eliminating seasonal effects to isolate underlying trend) windows. Third, time series decomposition separated the series into trend, seasonality, and remainder components using an additive mode, calculating seasonal factors by averaging detrended values across all years. Fourth, regression analysis implemented a multiple regression model using the Excel Analysis ToolPak, with time (months from series start) as a continuous predictor capturing warming trend, plus 11 seasonal dummy variables (February through December, with January as reference). Finally, the fitted regression model projected temperatures through November 2026, while summary statistics compared Early Era against Recent Era to quantify changes in average temperatures and snowmaking days.

3 Results: Trend, Seasonality, and Structure

The analysis revealed clear warming spanning 37 years. Average temperatures increased from 37.65°F in the Early Era to 39.89°F in the Recent Era, representing a 2.24°F rise. The trend appears linear with no structural breaks, and the 12-month moving average confirmed consistent upward trajectory. Seasonal patterns dominate temperature variation with a predictable annual cycle exhibiting 33°F amplitude from winter lows to summer peaks. Decomposition quantified seasonal factors: January coldest at 16.8°F below annual average, December at 16.0°F below, July and August peaks at 15.9°F and 16.2°F above average. The 3-month moving average dampened month-to-month noise while preserving seasonal rhythm, while the 12-month average transformed the volatile raw series into a smooth upward trajectory confirming warming rate. The decomposition remainder appeared small and randomly distributed with a standard deviation of 3.5°F, validating model effectiveness and supporting forecast reliability.

4 Regression and Forecasting Results

The multiple regression model achieved R-squared of 0.9295, explaining 92.95% of temperature variation through time trend and seasonal dummy variables. The model's F-statistic of 472.44 (p-value < 0.001) provides statistical evidence of significant predictive improvement. Standard error of 3.478°F indicated 95% of actual temperatures fall within approximately 7°F of predictions. The time coefficient estimated at 0.00795°F per month (p < 0.001, t-statistic = 6.15) quantifies warming rate, converting to 0.095°F per year or approximately 3.5°F over the 37-year period. This statistically significant positive trend confirms climate change affects this location. Seasonal dummy coefficients quantified temperature deviations from the January baseline: February 2.31°F warmer, March 8.23°F, April 14.67°F, May 23.12°F, June 32.63°F, July 32.93°F, August 33.14°F (peak), September 27.94°F, October 18.50°F, November 8.21°F, with December at 0.80°F (statistically insignificant, p=0.32). All other coefficients achieved p-values below 0.01. Forecasts extended through November 2026: December 2025 predicted at 24.46°F, March 2026 at 31.91°F, August 2026 peak at 56.86°F. These forecasts continue the

observed warming trend with seasonal adjustments, providing sufficient accuracy for multi-month planning decisions.

5 Actionable Recommendations and Reflection

The documented 2-3°F warming trend calls for strategic adjustments for Red River Ski Area. First, substantially expand snowmaking infrastructure. The analysis revealed 205 fewer days below 32°F in November and 317 fewer in December comparing Recent Era to Early Era, requiring increased reliance on artificial snow production. Management should prioritize energy-efficient equipment. Second, recalibrate marketing strategy. November temperatures average 1.8°F warmer in Recent Era, reducing reliable snowmaking windows. Marketing emphasis should shift toward December and January when temperatures remain sufficiently cold, positioning Red River as a mid-December through March destination, rather than Thanksgiving through April. Seasonal pass pricing could incentivize January-February visits when conditions are most reliable. Third, address long-term sustainability. If 0.095°F per year continues, another 3-4°F warming over the next 35-40 years brings cumulative warming to 6-7°F by the 2060s, threatening operations. Management should discuss scenario planning for diversification into summer recreation and evaluate expansion at higher elevations. This time series analysis demonstrated substantial value of quantitative analysis for business decisions. The analysis quantified warming rate, documented statistical significance, and translated abstract trends into concrete operational metrics like reduced snowmaking days. The regression framework separated legitimate trend from seasonal noise, enabling confident conclusions about climate dynamics. Most importantly, forecasting transformed historical analysis into forward-looking decision support with specific temperature predictions informing infrastructure investments, staffing, and marketing calendars.

Red River Climate Analysis (1989-2025)

38.80

Average Temperature (°F)

17.39

Minimum Temperature (°F)

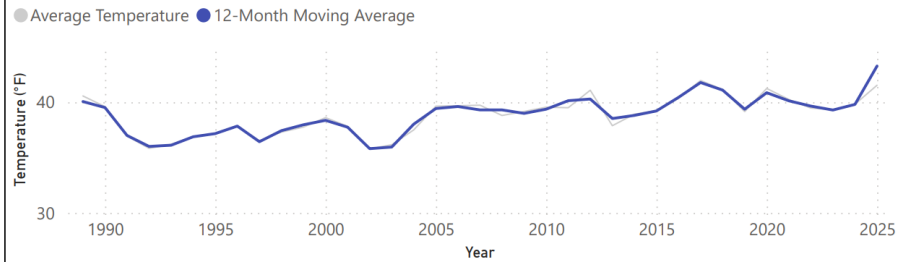
61.52

Maximum Temperature (°F)

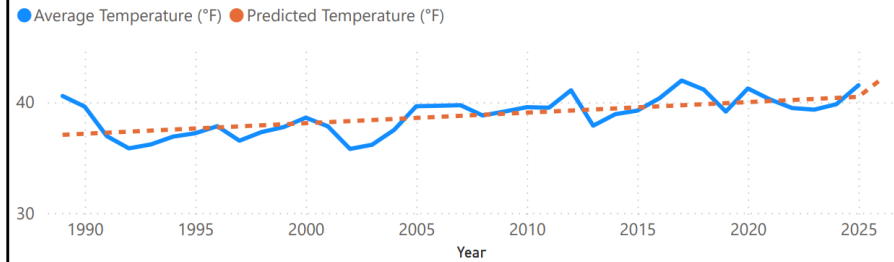
416

Months Analyzed

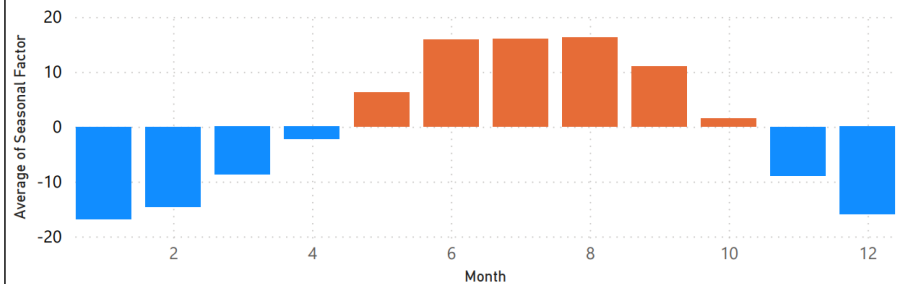
Monthly Temperature with 12-Month Moving Average



Temperature Trend & Forecast



Average of Seasonal Factor by Month



Era	Average Temperature (°F)	Minimum Temperature (°F)	Maximum Temperature (°F)
Recent Era	39.89	17.39	61.52
Early Era	37.65	17.68	59.58

KEY FINDING: Red River has warmed 2-3°F since 1989,

Recommendation: Red River Ski Area should invest in enhanced snowmaking capacity. December shows the largest warming (+2.7°F), reducing natural snowmaking days by 50%. Plan marketing around January (coldest, most reliable) and consider shoulder-season alternatives.